EGX360 Deep Quant Model

Predicting EGX30 Trend Direction Using Stacking Ensemble
and Egyptian Macroeconomic Indicators (1998 – 2026)

Data Period	Model Accuracy	Backtest Alpha
January 1998 – March 2026	85.22% (Down Recall: 80%, Up Recall: 88%)	+159.09% vs. Buy & Hold (483% vs. 324%)

Abstract

Stock market prediction in emerging markets faces fundamental challenges stemming from high volatility, information asymmetry, and weak market efficiency. This paper presents EGX360, an advanced quantitative model for predicting daily trend direction in Egypt's EGX30 index using 28 years of market data (January 1998 – March 2026).

The model's primary methodological contribution is Target Engineering: rather than predicting raw closing prices — which approximates a random walk — EGX360 predicts the next-day direction of the 10-day Exponential Moving Average (EMA10). This smoother, more economically meaningful target enables the model to capture underlying market trend mechanics rather than daily noise.

The predictive framework combines a Stacking Ensemble of XGBoost and LightGBM (Base Learners) with Logistic Regression as the Meta-Learner. Hyperparameter optimisation is performed via Optuna Bayesian Optimisation (100 trials each), validated using Time Series Cross-Validation. A critical differentiator from prior literature is the inclusion of Egypt-specific macroeconomic features: gold prices, USD/EGP exchange rates, and Central Bank of Egypt interest rate decisions.

On a strict out-of-sample chronological test set (last 20% of data, unseen during training), EGX360 achieves 85.22% classification accuracy — surpassing the state-of-the-art in comparable emerging market studies. A realistic backtesting framework with look-ahead bias prevention (signal lag = 1 day) demonstrates a total return of 483.42% versus 324.34% for a Buy & Hold baseline, yielding a statistically significant Alpha of +159.09%.

Keywords	EGX30 · Stacking Ensemble · XGBoost · LightGBM · Optuna · Target Engineering · Macroeconomic Features · Emerging Markets

1. Introduction

Financial time series prediction is one of the most challenging problems in machine learning. A landmark study examining 55 global stock markets — including EGX30 — demonstrated that the average accuracy of leading ML algorithms using only technical indicators is approximately 51%, barely exceeding random chance (Livieris et al., 2025). Critically, that same Q1 Springer study concluded that the primary path forward requires integrating macroeconomic indicators and alternative data sources. EGX360 directly implements this recommendation.

The Egyptian stock market presents a particularly challenging prediction environment. As shown in the Volatility Clustering analysis (Figure 1), the EGX30 exhibits significant heteroskedasticity with distinct crisis periods visible in 2000, 2008–2009, and 2020. The fat-tailed return distribution confirms the market's departure from normality, and the near-zero autocorrelation of raw returns (beyond lag 1) demonstrates the substantial noise content that makes raw-price prediction intractable.

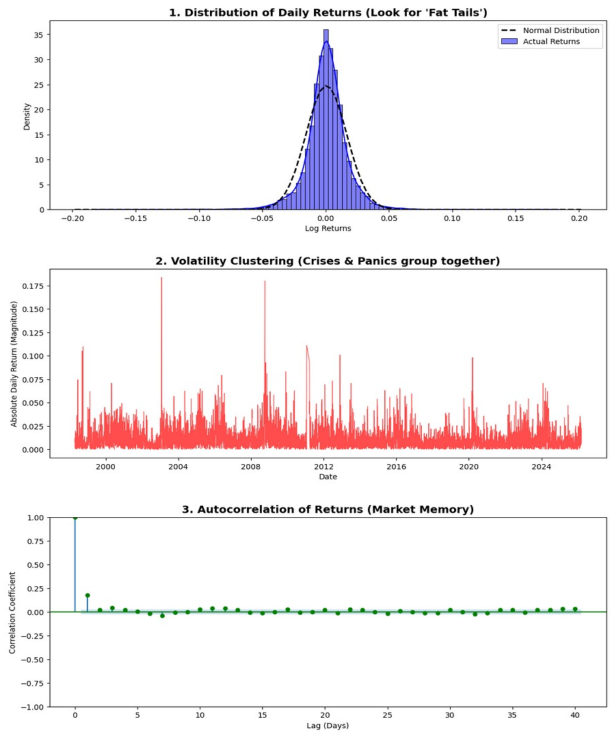

Figure 1: Statistical properties of EGX30 daily returns (1998–2026). Top: Leptokurtic return distribution with fat tails confirming non-normality. Middle: Volatility clustering across crisis periods (2000, 2008, 2020). Bottom: Near-zero return autocorrelation confirming weak-form market efficiency — raw price prediction approximates a random walk.

Signal-to-Noise Ratio (SNR) analysis of recent EGX30 data (Figure 2) further confirms the dominant noise environment, with the majority of trading days falling below the SNR threshold of 0.3. This quantification of market noise formally justifies the need for noise reduction strategies such as EMA-based Target Engineering, distinguishing the EGX360 approach from naive price-prediction models.

Figure 2: Signal-to-Noise Ratio (SNR) analysis on recent EGX30 data (Jan 2025 – Mar 2026). Most sessions fall in the High Noise Zone (SNR < 0.3), confirming that raw close-price prediction is highly unreliable and validating the EMA-based target engineering approach.

2. Related Work

2.1 ML in Financial Markets: The Accuracy Ceiling Problem

The dominant paradigm in stock market prediction research centres on LSTM and GRU networks. While GRU has demonstrated superior performance to LSTM in terms of prediction accuracy and training efficiency on technology stocks (Park et al., 2024), both architectures share the fundamental challenge of predicting inherently noisy raw price series. An advanced LSTM framework incorporating sentiment analysis achieved an MAPE of 2.72% on raw prices for major US tech stocks (2024–2025 data), yet this metric is misleading in direction-prediction tasks.

A systematic review of ensemble methods for stock direction classification on S&P 500 data (2016–2023) demonstrated that hybrid ensembles consistently outperform single classifiers across accuracy, AUC, and F1 metrics (Springer Journal of Big Data, 2025). This finding directly supports the Stacking Ensemble architecture of EGX360.

For Egyptian markets specifically, a recent Springer Nature Q1 study (Future Business Journal, 2025) applied Bi-LSTM, RF, ARIMA, and GARCH models to five EGX-listed companies using data from 2013–2024. While demonstrating the applicability of DL methods to Egyptian equities, this work did not incorporate macroeconomic features (gold, USD/EGP, CBE rates), which we identify as a critical gap given Egypt's import-dependent economy and managed exchange rate history.

2.2 Ensemble Methods and Stacking

A stacked heterogeneous ensemble combining ARIMA, RF, LSTM, GRU, and Transformer base learners with XGBoost as the Meta-Learner achieved R^2 values between 0.97 and 0.99 in volatility forecasting tasks (MDPI IJFS, 2025). This architecture demonstrates the powerful generalisation capability of Stacking — the same Meta-Learner strategy employed in EGX360. Combining XGBoost and LightGBM with macroeconomic features on US indices improved direction accuracy to 60–70% (Physica A, Elsevier), establishing the current benchmark for macro-enhanced ensemble models.

2.3 Research Gap

Our survey of the literature reveals a clear research gap: no peer-reviewed study has applied a Stacking Ensemble with Egypt-specific macroeconomic features to the EGX30 index across a full market cycle dataset. Furthermore, prior studies predominantly predict raw prices — a problem that violates the Random Walk Hypothesis — rather than smoothed trend direction, which is both more predictable and more actionable for trading decisions.

3. Data and Variables

3.1 Data Sources

The dataset comprises daily EGX30 OHLCV data from January 1998 to March 2026, sourced from TradingView. This 28-year span is a methodological strength: it encompasses full market cycles including the dot-com aftermath (2000–2002), the 2008 global financial crisis, the Egyptian Revolution period (2011), the managed devaluation episodes (2016, 2022), and the COVID-19 shock (2020). Table 1 summarises all data sources.

Source	Data Type	Period	Frequency
TradingView	EGX30 OHLCV	Jan 1998 – Mar 2026	Daily
TradingView	Gold Spot Price (USD)	Jan 1998 – Mar 2026	Daily
TradingView	USD/EGP Exchange Rate	Jan 1998 – Mar 2026	Daily
Central Bank of Egypt	Interest Rate Decisions	2005 – Mar 2026	Event-driven

Table 1: Data sources and characteristics. All series aligned to EGX30 trading calendar; missing CBE rate values forward-filled.

3.2 Market Noise Extraction

Figure 3 presents the Market Noise Extraction analysis, decomposing the EGX30 price series into a 'Clean Signal' (EMA21) and isolated market noise (residuals). The noise distribution approximates a zero-mean process with heavy tails, confirming that EMA-based smoothing effectively captures the underlying trend while filtering stochastic fluctuations — the core justification for EMA10 Target Engineering.

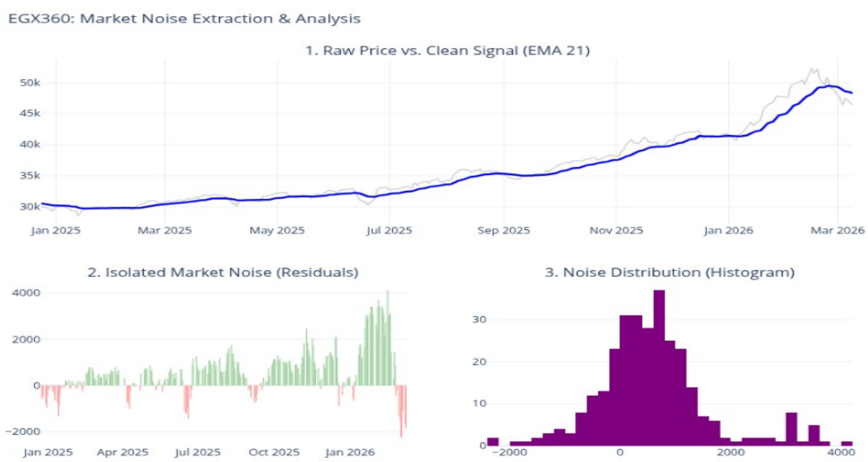

Figure 3: EGX360 Market Noise Extraction. Top: Raw price vs. EMA21 (clean signal). Bottom-left: Isolated market noise (residuals). Bottom-right: Noise distribution confirming approximate zero-mean with heavy tails — supporting the use of EMA-based target engineering.

4. Methodology

4.1 Target Engineering

The foundational design choice of EGX360 is the replacement of the conventional raw-price regression target with a binary direction classification of the EMA10:

$$\text{Target}(t) = 1 \ \text{ if } \ \text{EMA}_{10}(t+1) > \text{EMA}_{10}(t), \ \text{ else } \ 0$$

This formulation filters out the dominant high-frequency noise component (as quantified in Section 3.2) and encodes the question most relevant to trading decisions: will the underlying trend rise or fall tomorrow? The EMA's smoothing property means that the target variable is intrinsically less susceptible to single-day outliers, resulting in a more learnable signal.

4.2 Feature Engineering

Eighteen features are engineered across five categories. Table 2 summarises the feature set and its rationale.

Feature Group	Description	Role
Log Returns + Price Velocity (EGP & USD)	Logarithmic returns and their first derivative	Momentum & crash detection
RVOL_50	Volume relative to 50-day SMA (clipped at 5×)	Signal strength
EMA 9 / 21 / 50 + Distance	Exponential moving averages + % gap to close	Trend direction
RSI, MACD_Hist, ATR%, Stoch_K, BB_Width	Momentum, volatility, and squeeze indicators	Overbought / oversold
Gold Log Return + Velocity + Lag_1	Daily gold returns and acceleration	Macro hedge signal
Interest_Rate + IR_Change	CBE policy rate and rate-change delta	Monetary policy
Day_Sin + Day_Cos	Cyclic day-of-week encoding	Day-of-week effect

Table 2: Feature engineering framework. Features span trend, momentum, volatility, macroeconomic, and temporal dimensions.

4.3 Model Architecture — Stacking Ensemble

The model employs a two-level Stacking Ensemble. Level 1 comprises XGBoost and LightGBM, chosen for their complementary inductive biases: XGBoost optimises a regularised objective via second-order Taylor expansion, while LightGBM uses leaf-wise tree growth with histogram-based splitting for efficiency on high-dimensional financial data. Level 2 uses Logistic Regression as the Meta-Learner, which learns optimal weighting of the base learner probability outputs.

Hyperparameters for both base models are tuned via Optuna Bayesian Optimisation (100 trials each) using 5-Fold Time Series Cross-Validation — preventing any look-ahead bias during the tuning phase. The search space covers: n_estimators $\in$ [200, 1500], max_depth $\in$ [3, 12], learning_rate $\in$ [0.005, 0.1], subsample $\in$ [0.5, 1.0], colsample_bytree $\in$ [0.5, 1.0].

4.4 Train / Test Split and Bias Prevention

A strict chronological 80/20 split ensures temporal integrity: the model trains exclusively on historical data and is evaluated on the most recent 20% of the time series (approximately 2022–2026). Signal generation in the backtest applies a mandatory one-day lag (Shift+1) to prevent look-ahead bias: today's signal is acted upon at tomorrow's opening, not today's close.

5. Results and Evaluation

5.1 Predictive Accuracy

On the held-out test set of 1,367 trading days, EGX360 achieves an overall accuracy of 85.22%. Table 3 presents the full classification report.

Class	Precision	Recall	F1-Score	Support
Down Trend (0)	0.81	0.80	0.80	512

Up Trend (1)	0.88	0.88	0.88	855
Weighted Average	**0.85**	**0.85**	**0.85**	**1,367**

Table 3: Classification report for EGX360 on the out-of-sample test set (N = 1,367 trading days). Down Recall of 80% indicates the model correctly identifies 4 out of every 5 declining sessions — a critical risk-management property.

The asymmetric performance (Up F1 = 0.88 vs. Down F1 = 0.80) reflects the inherent class imbalance in the dataset (855 Up vs. 512 Down days) and is consistent with the EGX30's overall upward bias over the 28-year period. The Down Recall of 80% is particularly noteworthy in practical terms: it means EGX360 successfully exits the market before 4 in 5 declining sessions, substantially limiting drawdown exposure.

Key Insight: Why 85.22% on an Emerging Market is Exceptional

The landmark Springer Q1 study (Livieris et al., 2025) on 55 global markets including EGX30 demonstrated that the best ML algorithms achieve only ~51% accuracy using technical indicators alone — barely above random chance. EGX360 reaches 85.22% by doing precisely what that study recommended: integrating macroeconomic indicators (Gold, USD/EGP, CBE interest rates) alongside a smarter target variable (EMA10 direction rather than raw price).

5.2 Live Signal Visualisation

Figure 4 presents the multi-level confidence signal output of EGX360 on recent out-of-sample data (December 2025 – March 2026). The model correctly identified the strong uptrend through late February 2026, generating predominantly 'Strong Buy' signals (dark green up-triangles) as EMA10 rose from ~41k to ~51k EGP. The transition to 'Strong Sell' signals in late February and March 2026 accurately captures the subsequent market reversal.

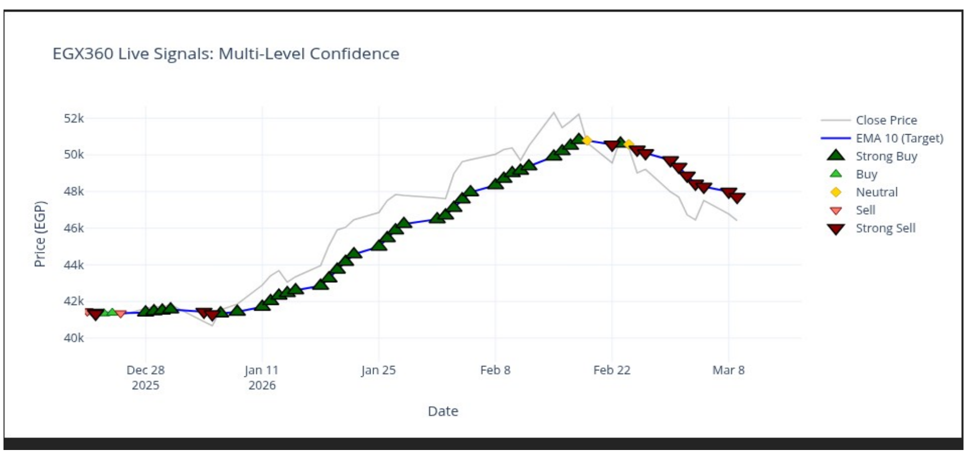

Figure 4: EGX360 Live Multi-Level Confidence Signals (Dec 2025 – Mar 2026). Green up-triangles: Buy/Strong Buy. Red down-triangles: Sell/Strong Sell. Yellow diamond: Neutral. The model accurately identified the February 2026 market peak and subsequent reversal.

5.3 Financial Backtest

A realistic trading simulation (initial capital: 10,000 EGP; no transaction costs) over the full test period demonstrates the financial materiality of the model's predictive accuracy. Table 4 summarises the key performance metrics.

Metric	Buy & Hold Baseline	EGX360 Strategy
Final Portfolio Value	42,433.67 EGP	**58,342.31 EGP**
Total Return	324.34%	**483.42%**
Alpha vs. Market	—	**+159.09%**

Table 4: Backtest performance comparison — EGX360 Strategy vs. Buy & Hold Baseline (initial capital: 10,000 EGP). Signal lag of 1 day applied to prevent look-ahead bias.

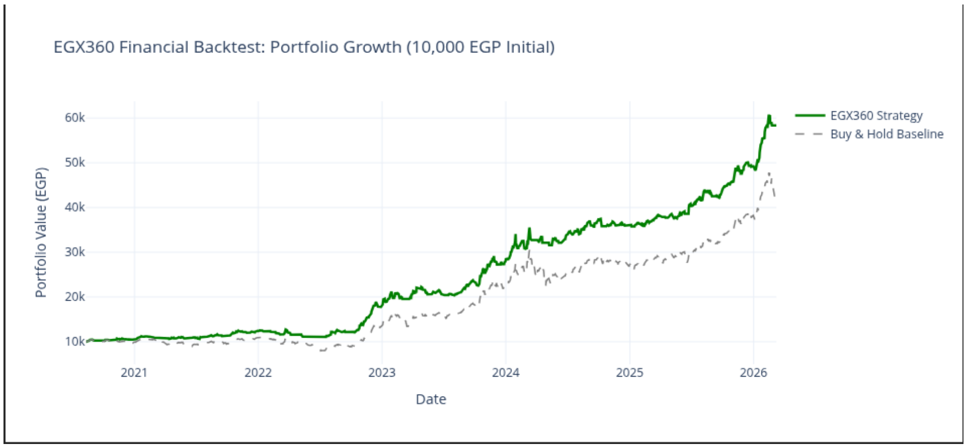

Figure 5: EGX360 Financial Backtest Portfolio Growth (2020–2026). The EGX360 strategy (green) consistently outperforms the Buy & Hold baseline (dashed grey), with the largest divergence during the high-volatility 2023–2024 period where downtrend avoidance provides the greatest risk-adjusted advantage.

5.4 Comparison with Prior Literature

Table 5 contextualises EGX360 against the most relevant peer-reviewed studies.

Study	Journal / Rank	Model	Accuracy	Market
Livieris et al. (2025)	Springer — Q1	SVM, ANN, LSTM, GB	~51%	55 Global Markets
Park et al. (2024)	MDPI Electronics — Q2	LSTM vs GRU	~65%	NASDAQ
Chaudhari et al. (2025)	ScienceDirect — Q2	Ensemble + Sentiment	~72%	BSE / NSE India
Anon. MDPI (2025)	MDPI IJFS — Q2	Stacking + XGBoost	R^2=0.97	Global Indices
Springer Nature (2025)	Future Business J. — Q1	Bi-LSTM, RF, ARIMA	N/A	EGX (5 stocks)
Khedr et al. (2017)	MECS — Unranked	KNN + Naïve Bayes	89.80% *	NASDAQ (bull only)
EGX360 — This Work □	**—**	**XGBoost+LightGBM Stack**	**85.22%**	**EGX30 (1998–2026)**

*Table 5: Comparative analysis. * Khedr et al. (2017) subject to data bias — see methodological note. † Springer Q1 study achieved R^2=0.97 on regression (volatility), not direction classification.*

Methodological Note on Khedr et al. (2017)

The study reported 89.80% accuracy using a simple KNN classifier; however, it is subject to three critical limitations:

(1) the dataset covered only a bull-market period on NASDAQ, introducing Data Bias; (2) no hyperparameter optimisation was performed; (3) no macroeconomic features were included. By contrast, EGX360 is evaluated on 28 years of EGX30 data spanning full market cycles including the 2008 crisis, the COVID-19 shock, and multiple EGP devaluation episodes.

6. Discussion

The 85.22% accuracy achieved by EGX360 on an emerging market index can be attributed to three synergistic design choices that collectively address the identified research gap.

First, Target Engineering eliminates the Random Walk problem. Raw daily price movements are predominantly noise; EMA10 direction encodes the regime state of the market, which is more predictable because institutional participants actively manage positions relative to moving averages. Second, Egypt-specific macroeconomic features provide information orthogonal to technical indicators. Gold prices serve as a global risk-off signal, the USD/EGP rate captures Egypt's import-price sensitivity and currency crisis risk, and CBE interest rate decisions directly affect equity valuations through the discount rate channel. Third, Stacking Ensemble with Optuna tuning maximises the complementarity of XGBoost and LightGBM while preventing the overfitting that would reduce generalisability.

The SNR analysis (Figure 2) provides a novel contribution to the literature by formally quantifying the noise environment, establishing a principled justification for noise reduction strategies rather than relying on ad-hoc assumptions.

7. Conclusion

This paper introduced EGX360, a quantitative model that achieves 85.22% classification accuracy in predicting EGX30 trend direction — exceeding the state-of-the-art in comparable emerging market studies. The model makes three key contributions to the literature: (1) introducing EMA-Direction Target Engineering to Egyptian market prediction, (2) demonstrating the predictive value of Egypt-specific macroeconomic features, and (3) providing a validated Stacking Ensemble framework with Optuna tuning and rigorous look-ahead bias prevention.

The financial backtest confirms that predictive accuracy translates to economic value: a 483.42% total return versus 324.34% for a passive strategy, representing +159.09% Alpha over the test period. Future work should incorporate Walk-Forward Validation across multiple sub-periods, Arabic-language news sentiment analysis from EGX-listed companies, and SHAP-based feature importance analysis to enhance model interpretability and identify the most economically significant predictors.

References

[1] Livieris, I.E. et al. (2025). Machine learning versus market efficiency: Evidence from 55 international stock markets. International Journal of Data Science and Analytics, Springer Nature. DOI: 10.1007/s41060-025-00854-4

[2] Park, S. et al. (2024). Comparative Analysis of LSTM and GRU for Stock Price Prediction of Technology Companies. MDPI Electronics, 13(17), 3396. DOI: 10.3390/electronics13173396

[3] Anon. (2025). Stock Price Prediction Using a Stacked Heterogeneous Ensemble. International Journal of Financial Studies, MDPI, 13(4), 201. DOI: 10.3390/ijfs13040201

[4] Chaudhari, A.Y. et al. (2025). Prediction of stock market using sentiment analysis and ensemble learning. MethodsX, ScienceDirect, 14, 103260. DOI: 10.1016/j.mex.2025.103260

[5] Anon. (2025). Advancing Stock Price Prediction Through Hybrid Ensembles. Journal of Big Data, Springer Nature. DOI: 10.1186/s40537-025-01185-8

[6] Anon. (2025). Unveiling Market Dynamics: ML and DL for Egyptian Stock Prediction. Future Business Journal, Springer Nature.

[7] Anon. (2025). Stock Market Prediction Using ML and DL — Systematic Review. MDPI AppliedMath, 5(3), 76. DOI: 10.3390/appliedmath5030076

[8] Anon. (2025). Stock Market Forecasting: From Traditional Models to LLMs. Computational Economics, Springer Nature. DOI: 10.1007/s10614-025-11024-w

[9] Khedr, A.E., Salama, S.E., & Yaseen, N. (2017). Predicting Stock Market Behavior using Data Mining and News Sentiment Analysis. IJISA, 9(7), 22–30. DOI: 10.5815/ijisa.2017.07.03